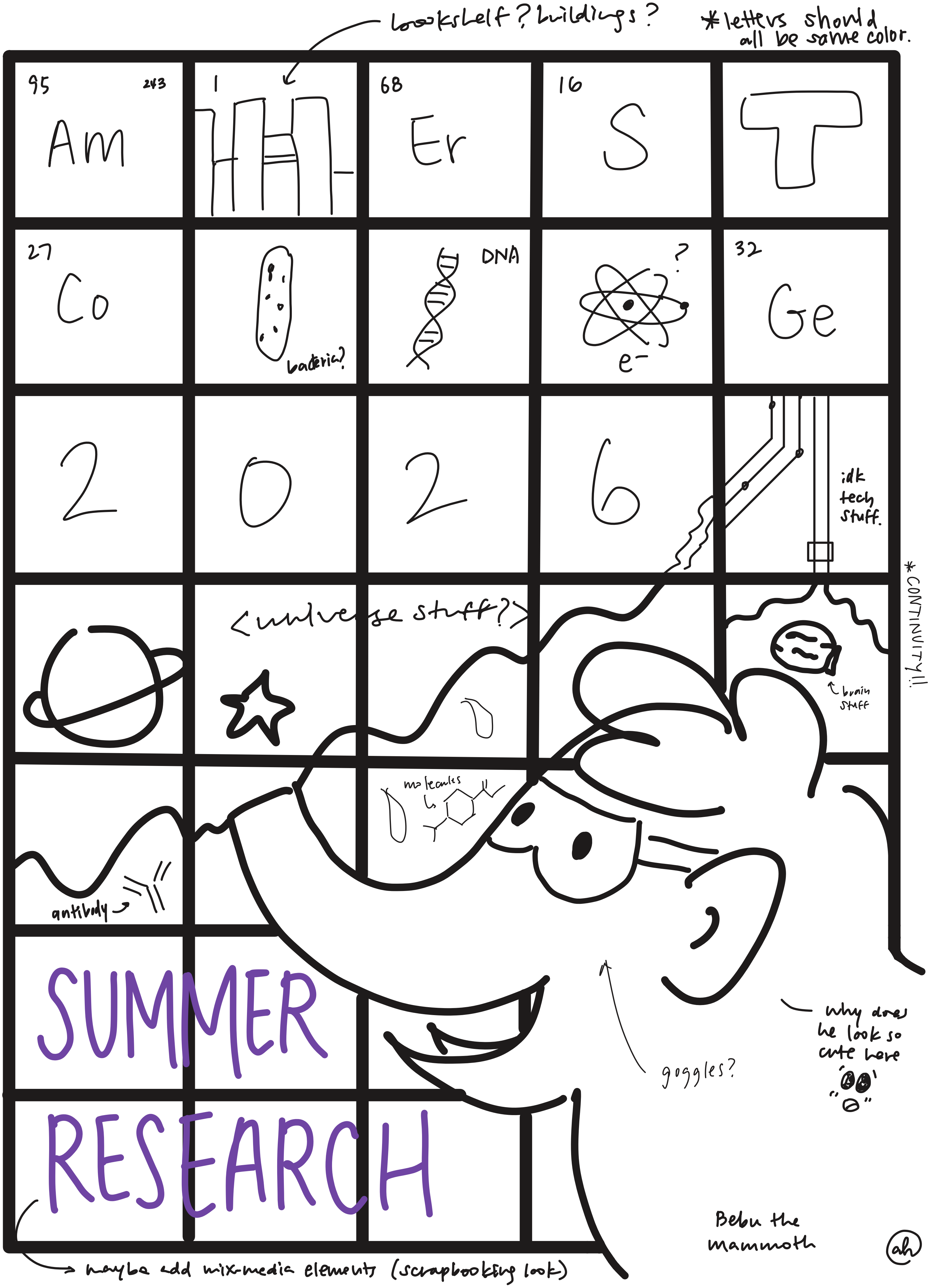
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brain stuff
molecules
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SUMMER
RESEARCH
goggles?
why does he look so cute here
Bebu the mammoth
maybe add mix-media elements (scrapbooking look)
ah

Amherst College Summer Research Abstract Book

2026

Amherst College Summer Research Abstract Book 2026

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This booklet showcases the research accomplished by Amherst College students in the summer of 2026. We hope it serves as a resource for future students interested in undergraduate scientific research.

Contents

Biochemistry and Biophysics

Optimizing Activity Assays for Alpha-Lytic Protease: A Foundation for Studying Protein Folding in Lysobacter enzymogenes

Emma Torres[1], Ellen Wolstenholme[2], Ryan Tak[2], Sheila Jaswal[1,2]

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Purpose. Purpose: Alpha-lytic protease (αLP) and beta-lytic metalloendopeptidase (βLM) are two enzymes produced by the soil bacteria Lysobacter enzymogenes. Its unique folding process makes both αLP and βLM interesting for industrial and agricultural use. The goal of our study is to develop a method for measuring αLP activity.

Methods. Methods: αLP activity was measured based on the breakdown of a synthetic substrate (Suc-Ala-Ala-Pro-Phe-pNA). Two measurement techniques were used: a UV/Vis spectrophotometer and a 96-well plate reader, they were compared for their sensitivity, sample volume requirements, and efficiency using serial dilutions of an archived αLP stock (2015). In addition to stock, αLP L. enzymogenes was cultured. Its growth was studied in two different types of nutrient media (Bachovchin's Media and Tryptone Soya Broth) to evaluate bacterial growth and natural αLP secretion, monitored by optical density and activity assays.

Results. Results: The plate reader produced clean and linear results. It required less volume of sample while allowing for multiple dilutions to be tested simultaneously, making it the more practical method going forward. A 1:100 dilution of the αLP stock yielded the most favorable and consistent readings. As for the cultivated L. enzymogenes bacterial growth differed between the two media, with lower optical density observed in Tryptone Soya Broth.

Conclusion. Conclusion: This study establishes a more efficient, plate reader-based protocol for measuring αLP activity and identifies key next steps for improving bacterial culture conditions to support future studies. These findings lay the groundwork for extending this method to βLM and for exploring alternative expression systems, such as Bacillus subtilis, ultimately enabling comparative research into how these two related proteases fold and stabilize.

Biology

Essential Roles of Six2 in Cranial Base Synchondrosis Development

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Purpose. The cranial base is a skeletal structure that contributes to skull growth and elongation. The cranial base synchondroses are cartilaginous joints and sites of endochondral growth, with the spheno-occipital synchondrosis (SOS) and intersphenoid synchondrosis (ISS) serving as major growth sites. Unlike long-bone growth plates, skeletal progenitor populations in the cranial base synchondroses remain poorly characterized. *Six2* is a transcription factor essential for the development of the kidney, limbs, and eyes. Single-cell RNA sequencing has identified *Six2* as highly expressed within the cranial base synchondroses. Global *Six2* knockout in mice results in premature SOS ossification and ISS loss, suggesting a crucial role for *Six2* in cranial base synchondrosis development. This project aimed to investigate the functional importance of *Six2* in postnatal cranial base synchondrosis development.

Methods. Using the *Cre-loxP* system, *Six2* was conditionally inactivated in postnatal chondrocytes using *Col2a1-CreER; Six2*$^{fl/fl}$*; R26R*^{tdTomato} (Col2a1-Six2-cKO) mice with tamoxifen injection at postnatal day 3 (P3). We also conditionally inactivated the chondrocyte master regulator *Sox9* in *Six2*$^{+}$ cells using *Six2-creER; Sox9*$^{fl/fl}$*; R26R*^{tdTomato} (Six2-Sox9-cKO) mice by injecting tamoxifen at P3.

Results. No significant differences in SOS or ISS length were observed between Col2a1-Six2-cKO and their controls. Lineage-tracing with the tdTomato reporter showed comparable labeling levels between Col2a1-Six2-cKO and control samples. In contrast, Six2-Sox9-cKO mice exhibited a significantly truncated ISS length associated with apparent premature ossification; however, interestingly, no significant difference in SOS length was observed. Increased tdTomato^{+} cells in Six2-Sox9-cKO mice suggest that *Sox9* loss in *Six2*$^{+}$ cells may affect chondrocyte differentiation.

Conclusion. Overall, these findings suggest that *Sox9* is required for *Six2*$^{+}$ cells to differentiate into chondrocytes and maintain postnatal cranial base synchondrosis particularly in ISS, while *Six2* itself is not required in these chondrocytes. Future studies will investigate the relationship between *Six2* and other chondrogenic regulatory genes to understand its role in cranial base synchondrosis development.

MYC Super Enhancer Activity in in T-ALL Requires TCF-1

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Purpose. Aberrant activation of MYC is a common oncogenic event across human cancers, including T-cell acute lymphoblastic leukemia (T-ALL). In approximately 5% of human T-ALL cases, MYC overexpression results from duplication and activation of N-ME, a distal Myc super-enhancer. We have developed a spontaneous mouse model of T-ALL driven by T-cell-specific stabilization of β-catenin combined with the loss of the transcription factor HEB (CATHEBΔ). This model faithfully recapitulates spontaneous N-ME super enhancer duplication and MYC overexpression during the early double positive (DP blast) stage of thymocyte development. We have also identified novel recruitment of TCF-1 to the N-ME super enhancer, and genetic ablation of TCF-1 from CATHEBΔ thymocytes completely prevents leukemogenesis. However, it remained unclear whether this protection resulted from disruption of thymocytes development, or whether TCF-1 has a key role in N-ME superenhancer activation via chromatin accessibility and remodeling.

Methods. To distinguish between these possibilities we characterized thymocytes development in CATHEBΔTCF1Δ mice using multiparameter flow cytometry with CD4, CD8, CD69, CCR7, CD25, TCRβ, TCF-1, and viability staining.

Results. Although total thymocytes numbers were significantly reduced consistent with the established role of TCF-1 in thymocyte development, CATHEBΔTCF1Δ thymocytes progressed to the DP blast stage that is targeted for leukemogenesis in CATHEBΔ mice. In addition, these mice exhibited significantly increased proportions of mature, post-DP T cells expressing CD69+ and CCR7+.

Conclusion. These findings demonstrate that loss of leukemogenesis in CATHEBΔTCF1Δ mice cannot be explained solely by a developmental block at the DP stage. Instead, they support a direct role for TCF-1 in promoting leukemic transformation, consistent with its proposed function in activating the N-Me super-enhancer and driving MYC-dependent T-ALL.

Phenocopying ΔacrZ through Chromosomal Mutations in the AcrB Efflux Pump

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Purpose. Antibiotics are essential in treating bacterial infections. However, some bacteria have developed the ability to survive in the presence of antibiotics. This antibiotic resistance is a threat to individual and public health. In Escherichia coli, one such mechanism is the AcrAB-TolC multidrug efflux pump, a protein pump in the bacteria's cell membrane that pumps antibiotics out of the cell. The deletion of the small protein AcrZ, which is associated with the pump, changes susceptibility to some antibiotics. To better understand the specific role of AcrZ, its relation to AcrB, and its interaction with certain classes of antibiotics, we sought to phenocopy the impact of an acrZ deletion through point mutations corresponding to residues 563 and 676 in the amino acid sequence for AcrB.

Methods. We first inserted the cat-sacB cassette, which confers chloramphenicol resistance and sucrose sensitivity, onto the E. coli chromosome using lambda red recombineering and allelic exchange. The cassette allowed us to test for proper chromosomal insertion using the cat-sacB phenotype, with the goal of later replacing cat-sacB with our mutants of interest.

Results. While both allelic exchange and lambda red recombineering yielded chloramphenicol resistant strains, neither method produced sucrose-sensitive strains, indicating improper integration of cat-sacB or sacB mutation.

Conclusion. Further troubleshooting and sequencing of the strains are required before our mutants of interest can be inserted.

Chemistry

Exploring the Photophysics of Mn2+ Dopant in ZnSe QD System

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Purpose. To learn more about the Mn:ZnSe system, we aim to produce a series of Mn:ZnSe QDs with different doping amounts to observe their characteristics and differences. We also want to adjust the synthesis procedure to tune the size and produce uniform dots. We aim to also optimize the EPR settings as Mn has a very strong signal. To further understand the spin of Mn2+, we will do CW-EPR to characterize the spin of Mn2+ at varying concentrations in ZnSe QDs.

Methods. Mn:ZnSe quantum dots are synthesized in an air free environment, with the degassing and heating times modified to tune QD size. UV-Vis spectroscopy and emission are used to observe absorption, emission, and relative quantum yield. Electron paramagnetic resonance (EPR) spectroscopy is performed to observe the spin of Mn2+. To determine size and doping percent of the QDs, transmission electron microscopy and ICP mass spectroscopy are performed.

Results. The results of this study suggest that the Mn:ZnSe QDs synthesized express similar characteristics to those in literature. Samples with higher inputs of Mn have lower lifetimes and higher emission values, as expected. The EPR spectra of samples also display the six peaks expected. However, with very high inputs of Mn, samples experience signal disruption, leading to lower emission, or Mn-Mn coupling, leading to a different EPR signal.

Conclusion. The observations from this study demonstrate potential in synthesizing a broader series of Mn:ZnSe QDs with different Mn2+ doping percentages. Beyond just characterization, we also aim to determine the electron transfer rate, through methods such as quenching or shelling, and photophysics of Mn:ZnSe QDs.

Investigating Localized Surface Plasmon Resonance Effects in Hybrid AuNP-TADF Thin Films

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[3] Department of Chemistry, University of Puerto Rico at Cayey, Cayey, PR

Purpose. Thermally activated delayed fluorescence (TADF) materials exhibit temperature-dependent photophysical changes, as thermal energy facilitates reverse intersystem crossing from the triplet to the singlet excited states, giving rise to delayed fluorescence. Gold nanoparticles (AuNPs) exhibit localized surface plasmon resonance (LSPR), through which absorbed light can generate localized thermal effects. This study investigates how LSPR from AuNPs influences the photophysical behavior of the TADF emitter 2,4,5,6-tetra(9H-carbazol-9-yl)isophthalonitrile (4CzIPN) in polymer thin films, with emphasis on changes in its delayed fluorescence lifetime and its potential as a probe of local temperature changes near AuNPs.

Methods. Polymer thin films containing poly(methyl methacrylate) (PMMA), 4CzIPN, and AuNPs were prepared by spin coating and drop casting and characterized using ultraviolet-visible (UV-Vis) absorption and steady-state emission spectroscopy. Temperature-dependent fluorescence lifetime measurements were used to determine how the lifetime of 4CzIPN changes with temperature. Hybrid AuNP-4CzIPN films were further examined using dark-field microscopy, fluorescence lifetime imaging microscopy (FLIM), and transient absorption spectroscopy to investigate their photophysical response to plasmon excitation.

Results. The delayed fluorescence lifetime of 4CzIPN in polymer thin films decreased with increasing temperature, establishing a relationship between the photophysical response of 4CzIPN and thermal changes in its environment. In hybrid thin films containing both 4CzIPN and AuNPs, a decrease in fluorescence lifetime was also observed following plasmon excitation, consistent with a possible contribution from LSPR-generated heating to the observed lifetime decrease. However, the observed changes in lifetime cannot yet be attributed exclusively to plasmonic heating, as other interactions between AuNPs and 4CzIPN, including quenching, may also contribute to the measured response.

Conclusion. These results show the potential for using the temperature-dependent delayed fluorescence lifetime of 4CzIPN to probe local temperature changes in hybrid AuNP-TADF thin films. Further work is needed to separate LSPR-generated heating from other AuNP-4CzIPN interactions, such as quenching, and determine whether TADF emitters can be used as local temperature probes near plasmonic nanoparticles.

Kinetics of Solid-State Polymerization of H6TNAP

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Purpose. 11,11,12,12-Tetracyanonaphtho-2,6-quinodimethane, which we abbreviate as H6TNAP, is one of few known molecules to polymerize in the solid state. Unlike polymerization in the liquid or gas state, solid state polymerization provides a high degree of control over polymerization reactions. These controlled polymer syntheses have applications in materials science, specifically for producing copolymers with the capability to stabilize donor-acceptor interfaces in electronic materials, like semiconductors and photovolatics. However, due to the rarity of solid-state polymerization in organic materials, much is still unknown about their kinetics. Our project seeks to explore the kinetics of solid-state polymerization in an organic crystalline substance, focusing on how the polymerization propagates and the factors that explain why.

Methods. H6TNAP Polymerization occurs under conditions of high heat and UV-Vis light. We created these conditions using a hot plate set between 200-220ºC and a photoreactor, 3D printed to keep an LED light ˜2 cm away from the sample. We created samples on 40 mm glass cover slips, isolating 5-7 H6TNAP crystals on the center of each and sealing with another cover slip kept 0.06mm apart using one layer of Kapton tape. Because the H6TNAP monomer is red and its polymer is colorless, we collected data using a brightfield microscope, which accurately indicated the polymerization using average intensity measurements. Images were captured every 5 minutes and analyzed using Zen Imaging software to create time series for each trial.

Results. We observed both linear and quadratic average intensity time series. Linear time series were most often observed in data sets with less time points, while quadratic time series were observed in longer data sets, like those with larger crystals or lower temperature trials. Via image analysis, we observed that perimeter to area ratio increased over time and that core intensity increased over time as well.

Conclusion. Given that shorter time series yielded linear data, we hypothesized that the standard polymerization kinetics of our organic crystals are linear by area. Under this assumption, our image analysis observations allowed us to identify factors that explained the quadratic average intensity time series, namely an increasing perimeter:area ratio and 3D polymerization, which we hypothesized to cause slight increases in polymerization rate over time in larger crystals or lower temperature trials. Currently, we do not have adequate data to necessarily prove our hypotheses. Our next steps are to improve methods as to run trials more efficiently and with less miscellaneous error.

Microwave Rotational Spectroscopy: Determining the Structure of 3-Chloro-2,3,3-Trifluoropropene Monomer and its Complexes with Argon and Acetylene

Amaia C. Williams[1], Sophia Zhuang[1], Helen O. Leung[1], Mark D. Marshall[1]

[1] Department of Chemistry, Amherst College, Amherst, MA

Purpose. Although weaker than chemical bonds, intermolecular interactions are essential for human life. The Leung and Marshall lab examines such interactions through exploring the configurations of halogenated ethenes and propenes. As a part of the lab's study of halopropenes, we seek to determine the structure of 3-chloro-2,3,3-trifluoropropene (3Cl233TFP).

Methods. Gaussian16 was used to generate predicted structures of the monomer and its isotopologues. From the theoretical structure, rotational constants were obtained to generate a predicted spectrum. An experimental spectrum was obtained through a broadband, chirped-pulse Fourier transform microwave spectrometer. Comparing the two spectra in PGOPHER, the predicted rotational constants were adjusted to match experimental data. Experimental rotational constants for the most abundant species (and its isotopologues) were used to determine an experimental structure through Kraitchman coordinates and STRFIT calculations. Similar methods were used for 3Cl233TFP's complex with argon and acetylene. For the acetylene complex, basis set superposition error (BSSE) and zero point energy (ZPE) corrections were applied to determine the lowest energy configuration.

Results. The potential energy scan of the 3Cl233TFP monomer yielded two unique rotamers, from which the lowest energy configuration was determined. The spectra of both rotamers and their Cl-37 isotopologues were assigned to obtain rotational constants; C-13 isotopologues were only assigned for the lowest energy rotamer. An experimental structure for the monomer was obtained. An energy scan of argon around 3Cl233TFP's rotamers resulted in 3 distinct minima for the lower energy configuration and 1 distinct minima for the higher energy configuration. Experimental rotational constants were only determined for the lowest energy Argon complex ($r = 4.0132$, $\theta = 134.7254$, $\Phi = 65.6724$) and its Cl-37 isotopologue. An experimental structure of the Ar-3Cl233TFP complex was determined. An energy scan of acetylene around the lowest energy rotamer resulted in 3 distinct minima. BSSE and ZPE corrections were applied to identify the absolute minima at ($r = 4.1781$, $\theta = 71.8738$, $\Phi = 234.7507$).

Conclusion. Theoretical predictions of the 3Cl233TFP monomer and its complexes were adequate, as experimental and predicted structures yielded similar results. Further explorations of the HCCH-3Cl233TFP complex are ongoing to determine the experimental structure of the HCHH-3Cl233TFP structure.

Microwaving Molecules Until They Tell Us Their Secrets: Spectroscopic Characterization of a Halopropene

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Purpose. In quantum systems such as molecules and their complexes, angular momentum is quantized. This quantization yields distinct transitions that can be measured and exploited to determine the precise structure of the molecule and its complexes with other atoms. By determining the structure of such a system, a fuller understanding of intermolecular forces can be advanced. In this project, we use microwave spectroscopy to characterize the compound 3-chloro-1,1,3,3-tetrafluropropene and its complex with argon.

Methods. An experimental spectrum was measured using Chirped-Pulse Fourier-Transform Rotational Microwave (CP-FTMW) Spectroscopy. To predict discrete transitions, the system was modeled with the quantum chemistry software Gaussian16. Subsequently, the theoretical spectrum was simulated and assigned to the experimental spectrum using PGOPHER, and revised structural coordinates were calculated where possible.

Results. A dihedral scan of the monomer revealed two enantiomeric rotamers, which have identical spectroscopic constants. We then matched peaks for the Cl35 and Cl37 isotopologues of the monomer. We fit rotational constants, distortion constants, and quadrupole coupling constants to the best of their ability based on the transitions assigned. We then scanned Argon around the monomer for the argon complex, and we took the lowest energy configuration to assign transitions for the Cl35 and Cl37 isotopologues of the argon complex.

Conclusion. The theory we used provided an estimate that lay within 10% of the value of our assigned constants, producing a relatively accurate theoretical structure. In the future, we want to look at the acetylene complex with the monomer to find out more about the structure of the molecule and its intermolecular interactions.

Optimizing Synthesis of InP/MnZnS Quantum Dots

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Purpose. This study aimed to synthesize and characterize manganese (Mn)-doped core/shell indium phosphide/zinc sulfide (InP/ZnS) quantum dots (QDs) with blue and green emission. Mn doping introduces additional optical properties to semiconductor QDs, making these materials of interest for applications in optoelectronics. A major challenge encountered during synthesis was the formation of phosphate byproducts, which interfered with the characterization of the desired QDs. We focused on modifying the synthesis conditions to improve QD formation while minimizing the amount of phosphates.

Methods. InP/ZnS QDs were synthesized using a hot-injection method, with modifications made to reaction conditions between syntheses to optimize QD formation. Distinct samples were synthesized in the green-emitting series. Absorption and emission spectroscopy were used to characterize the QDs and determine their emission behavior, while transmission electron microscopy (TEM) was used to examine the morphology and size of QDs. Changes made between syntheses were assessed based on their effects on product formation and characterization.

Results. Sample MG02 served as a control sample for comparing subsequent modifications and showed significant phosphate formation. Repeating the synthesis with a degassed petrolatum jelly (PJ):octadecene (ODE) mixture and increased stirring speed in sample MG03 did not substantially resolve phosphate formation. Changing the inert atmosphere from house nitrogen to Grade 5 argon in sample MG04 resulted in a decrease in the amount of phosphate observed. Further modifications in sample JHK22, based on previously successful synthesis protocols, including consolidation of the precursor injections, use of a fresh tris(diethylamino)phosphine source, and stricter control of heating times, resulted in a further decrease in phosphate formation, although phosphate byproducts remained present.

Conclusion. Synthesis modifications for the InP/ZnS QDs procedure demonstrated that reaction conditions, including inert atmosphere, precursor amount, injection strategy, and heating control, can influence phosphate byproduct formation. The decrease in phosphate observed following the use of Grade 5 argon and subsequent protocol modifications suggests that these changes improved the synthesis conditions, although further optimization is necessary to eliminate phosphate formation and obtain consistently characterizable QDs. Uv-Vis/photoluminescence spectroscopy and TEM characterization provided methods for identifying conditions of successful Mn-doped and undoped InP/ZnS QDs.

Structure Determination of 2-Chloro-3,3,3-Trifluoropropene and its Complex with Argon via Microwave Rotational Spectroscopy

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Purpose. Intermolecular forces are a fundamental aspect of chemistry at all levels, and their nuances can be elucidated via microwave rotational spectroscopy. Upon absorption or emission of specific wavelengths of light, molecules transition between quantized rotational states, which are recorded; the spectrum and the resulting structural information are then used to determine the precise structure of the molecule or complex. By applying these techniques to 2-chloro-3,3,3-trifluoropropene and its complex with argon, we further the ongoing halopropene research in the Leung-Marshall lab.

Methods. A 2/3% sample of 2-chloro-3,3,3-trifluoropropene mixed with argon was analyzed using chirped-pulse Fourier transform microwave spectroscopy from 2 to 18 GHz. Ab initio calculations were performed using the MP2 level of theory using Gaussian16 software to predict rotational constants, which were then used in Western's PGopher software to obtain experimental constants. Bond lengths and angles were then calculated using Kisiel's STRFIT, yielding cartesian coordinates for the atoms.

Results. A potential energy scan of the monomer identified a lowest-energy conformation, and the transitions of 5 isotopologues were assigned in the experimental spectrum, allowing determination of 5 sets of rotational constants. Using STRFIT, 6 geometric parameters were fitted to the 15 rotational constants to obtain an experimental structure for the monomer. Scans of the argon complex yielded one minimum, which was identified and fitted on the experimental spectrum, along with its Cl-37 isotopologue. The 6 new constants and the results from the monomer fit were then used to determine the position of the argon atom in the complex, using the same methods.

Conclusion. The theoretical values proved to be a close prediction for our monomer and argon complex as the differences between predicted and experimental rotational constants were small. We identified the energy minima and obtained experimental structures for the monomer and its complex with argon. Next steps include performing similar analytical techniques on the monomer complex with acetylene.

Geology

Was Plate Tectonics Active During the Archean Eon? A Paleomagnetic Approach

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Purpose. The theory of plate tectonics is the central paradigm of modern geology, as it provides a framework for understanding the history and morphology of continental and oceanic crust, as well as the distribution of volcanoes and seismic zones on the modern Earth. While there is evidence that plate tectonics has been active throughout the Phanerozoic, there is no consensus about when it began, nor what characterized the lithospheric regime that preceded it. Determining the nature and timing of the onset of plate tectonics is essential to understanding ancient climate and the development of life, as tectonic processes play an essential role in the modern carbon cycle.

Methods. This study investigates ancient tectonics through paleomagnetic analysis of ˜3.1 Ga volcanic rocks of the Sholl Terrane, collected from the Northwest Pilbara Craton in Australia. The orientation of the ancient magnetic field was determined by measuring the magnetic moment of oriented rock samples in a 2G Enterprises DC-SQuID Superconducting Rock Magnetometer. Through progressive thermal demagnetization past the blocking temperatures of the dominant ferromagnetic mineral phases, the characteristic remanent magnetism (ChRM) component was isolated.

Results. ChRM orientations will be used to define a paleomagnetic pole for the Northwest Pilbara Craton at ˜3.1 Ga. This will be compared to paleomagnetic poles from adjacent units to quantify the extent of apparent polar wander (APW) and evaluate the possibility of mobile-lid tectonics in the Archean.

Conclusion. Determining the nature and timing of the onset of plate tectonics is essential to understanding ancient climate and the development of life, as tectonic processes play an essential role in the modern carbon cycle. This research seeks to shed light on the early tectonic history of the Earth, helping to explain unanswered questions about our planet's physical and biological history.

Mathematics and Statistics

Generalized Bracket Rules for $B(\infty)$ Crystals

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Abstract: The crystal $B(\infty)$ is an infinite-dimensional crystal basis which is associated with the lower half of a quantum group. It acts as a combinatorial model, encoding information about its semisimple Lie Algebra and finite-dimensional representations.

Lusztig, Berenstein, and Zelevinsky have developed an algorithm to realize the whole crystal for a given reduced expression for the longest element w_0 in the Weyl group. This algorithm has to do with taking a PBW datum with respect to the chosen reduced word, which becomes a vertex of the crystal, and applying the crystal operator, which becomes an edge of the crystal. Tingley, Salisbury, and Schultze found a simpler realization where the crystal operators are represented by a bracketing rule. This bracketing rule was restricted to one of the four classical Lie types, type A_n as well as for a special case of word called a braidless word in the other three classical Lie types, B_n, C_n, and D_n.

We generalized the above bracket rule beyond braidless words for specific reduced expressions. This extends the use of this efficient algorithm above as it eliminates the issue of the restriction to braidless words only allowing for a bracket rule for the ordinary crystal operators or for the *-operators, which are reversed due to Kashiwara's involution, but not for both types of operators simultaneously. With our result, there is a simple combinatorial model of the algorithm in the case of both the ordinary crystal operators and the *-operators on a single realization of $B(\infty)$ in types B_n, C_n, and D_n.

This material is based upon work supported by the National Science Foundation under Grant No. DMS-1929284 while the authors were in residence at the Institute for Computational and Experimental Research in Mathematics in Providence, RI, during the Summer@ICERM 2026: Pure and Applied Mathematical Models program.

Singular Fibers of Bending Systems on Hexagon Spaces

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The moduli space $\mathcal{P}(r_1, \ldots, r_n)$ of closed n-gons in $\mathbb{R}^3$ with fixed side lengths $r_i > 0$ modulo rigid rotations is a smooth $2(n-3)$-dimensional symplectic manifold for generic side lengths. Hamiltonian actions generated by diagonal bending flows yield completely integrable systems on these spaces, indexed by triangulations of the n-gon. For $n = 6$, there are two combinatorially distinct integrable systems: the caterpillar (fan) system $F_{\text{cat}} = (l_{12}, l_{56}, l_{123})$ and the symmetric (star) system $F_{\text{sym}} = (l_{12}, l_{34}, l_{56})$. The image of the moment map $F : \mathcal{P}(r) \to \mathbb{R}^3$ is a 3D convex moment polytope.

A vertex of the moment polytope is Delzant if it is simple and smooth. At such points, the action is locally toric and the fiber is a single point. When a vertex is non-Delzant, the corresponding singular fiber can be a higher-dimensional manifold. We classify all conditions on generic side length vectors under which non-Delzant vertices appear for both bending systems and explicitly compute the topology of their singular fibers.

In the symmetric system, a non-Delzant vertex occurs if and only if two adjacent side lengths are equal (e.g., $r_1 = r_2$) and the others are at their extremes. Reconstruction reveals that every non-Delzant vertex in F_{sym} has a singular fiber diffeomorphic to S^2. In the caterpillar system, non-Delzant vertices arise either from an edge equality as above or when $z = l_{123} = 0$, causing the hexagon to decouple into two independent closed triangles $T_1 = (r_1, r_2, r_3)$ and $T_2 = (r_4, r_5, r_6)$. We prove that $F_{\text{cat}}^{-1}(v) \cong SO(3)$ if and only if both triangles strictly satisfy the triangle inequalities in this latter case. In all other generic non-Delzant cases, including degenerate triangles, the fiber is diffeomorphic to S^2. This completely classifies the topology of singular fibers over non-Delzant vertices in generic hexagon spaces.

An Algebraic Encoding of Bipartite R.E.C.s

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Bipartite regular edge colorings (R.E.C.s) are edge-colorings of a balanced bipartite graph $G = (V_1 \sqcup V_2, E)$ with $|V_1| = |V_2| = m$, using n colors, such that every vertex is incident to exactly one edge of each color. Classically, R.E.C.s can be represented as sums of permutation matrices. While this representation is useful computationally, it does not directly expose how different color classes interact. The union of two color classes decomposes into alternating cycles. We give an algebraic description of bipartite R.E.C.s via compatible families of permutations/derangements, and prove that this encoding loses no information up to relabeling of one bipartition.

Each perfect matching can be encoded by an involution $\tau_i \in \mathrm{Sym}(V)$ that exchanges the two parts V_1 and V_2. Composing two colors i and j gives a permutation of V_1,

$$\delta_{ij} = (\tau_j \circ \tau_i)|_{V_1}.$$

For $i \neq j$, this permutation is fixed-point-free; hence δ_{ij} is a derangement. These derangements are not independent: they satisfy the gluing relation

$$\delta_{k\ell} \circ \delta_{ak} = \delta_{a\ell}.$$

Thus, the interaction of the color classes is encoded by a compatible family of permutations on V_1.

A family $D = (\delta_{k\ell})_{k,\ell\in[n]}$ is compatible if it satisfies the identities $\delta_{kk} = \mathrm{id}_{V_1}$, the gluing condition, and fixed-point-freeness for every $k \neq \ell$. Conversely, every compatible family comes from an abstract R.E.C. by choosing any bijection $\tau_1 : V_1 \to V_2$ and defining the involutions τ_ℓ. Realization is unique up to relabeling V_2.

Therefore, each classical R.E.C. determines a compatible family, and conversely. In particular,

$$\mathcal{G}_n/\!\sim\ \cong\ \mathcal{F}_n/\!\sim\ \cong\ \mathcal{D}_n,$$

where $\mathcal{G}_n$ is the collection of classical bipartite R.E.C.s with n colors, $\mathcal{F}_n$ is the collection of abstract R.E.C.s encoded by involutions, and $\mathcal{D}_n$ is the collection of compatible families of derangements.

Medicine

Multitargeting CYP3A4, PI3Kα, and CDK4/6 Overcomes Hormone Therapy Resistance in Mutant PI3Kα ER+/HER2− Breast Cancer

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Purpose. About 40% of patients with estrogen receptor-positive HER2 negative (ER+ HER2-) metastatic breast cancer (MBC) exhibit hormone therapy resistance (HTR) driven by mutant PI3Kα, limiting the effectiveness of endocrine therapies such as the selective estrogen receptor degrader (SERD) fulvestrant (F). Mutant PI3K HTR is associated with an increase in the arachidonic acid (AA) precursor of cancer promoting -6 epoxy polyunsaturated fatty acids (-6 EpFAs) and upregulation of the enzyme CYP3A4 that synthesizes EpFAs from AA. In the first part of this study, we measured levels of various proteins, including CYP3A4, involved in breast cancer signaling pathways, to find which proteins are upregulated in HTR cells. In the second part of this study, we tested different drug combinations in a synergy study to find out how to overcome hormone therapy resistance.

Methods. F-selected MCF-7AC1 clonal cell lines were screened and the rate of CYP3A4 upregulation was measured using a western blot. Next, we treated model HTR cells with the mutant PI3Kα inhibitor tersolisib (T) and the CYP3A4 inhibitor C5F2-HCB and measured Akt, ERK, and S6 kinase phosphorylation. We also treated model HTR cells with the cell cycle inhibitor abemaciclib (A) and C5F2-HCB. Here, we measured Cyclin E1 and CDK6 levels. Lastly, we tested the combinations A+T, A+T+F, and A+T+F+C5F2-HCB in a MTT viability assay. The computer program CompuSyn produced combination index values (CI values) at different effective doses.

Results. CYP3A4 was upregulated in 22.2% of cells. Treatment with T significantly decreased Akt phosphorylation. Treatment with C5F2-HCB increased Akt and decreased ERK and S6 Kinase phosphorylation. Surprisingly, we found that treatment with A increased Cyclin E1 Levels and treatment with C5F2-HCB decreased CDK6 levels. C5F2-HCB synergized with a combination of the mutant PI3Kα inhibitor T, the CDKi A, and F (CI = 0.27). When A, T, F, and C5F2-HCB are combined, there was higher cell inhibition for all drug concentrations compared to when C5F2-HCB was administered alone. The quadruple combination showed the best synergy, with all CI values significantly less than one.

Conclusion. C5F2-HCB, a compound our lab has made previously that inhibits CYP3A4, could be used in multitargeted drug therapies. Treatment with C5F2-HCB decreased CDK6 levels suggesting that C5F2-HCB may overcome resistance to CDKis such as P and A in HTR mutant PI3Kα cells. Finally, a combination of 4 drugs—C5F2-HCB (CYP3A4i), abemaciclib (CDKi), fulvestrant, and tersolisib (mutant PI3Kαi)—show strong synergy. This means multitargeted inhibition of CYP3A4, CDKs, and mutant PI3Kα not only overcomes HTR, but suggests decreased toxicity and increased treatment efficacy for patients in the clinic. This promising combination supports its further evaluation in preclinical models.

Removing Barriers to Alcohol Use Disorder Treatment: Public Interest in Over-the-Counter Naltrexone

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Purpose. This study aims to explore public interest in over-the-counter (OTC) naltrexone and assess whether this approach may serve as a feasible strategy to expand access to alcohol use disorder treatment. Individuals' awareness, attitudes, and concerns toward OTC naltrexone have not yet been systematically studied, so this information is needed to guide future policy and education efforts to increase access to safe, low-barrier, and effective treatment for AUD.

Methods. An over-the-phone survey assessed alcohol use history, awareness of naltrexone, willingness to use naltrexone if it was available OTC, perceived benefits and concerns related to OTC use, and preferred sources of guidance if such a medication were available without a prescription. Inclusion criteria were English-speaking, 18+ adults with self-reported alcohol use at any level.

Results. The results of this retrospective study indicate statistically significant positive effects of mammoth stuffed animals on student well-being. Analysis of the quantitative data from surveys revealed a significant reduction in students' reported stress levels ($p < 0.001$) and a notable increase in self-esteem ($p < 0.05$) when interacting with these plush companions. Moreover, qualitative insights obtained from interviews and observations highlighted the presence of mammoth stuffed animals as a significant factor contributing to a sense of security and emotional support among students. The findings emphasize the potential efficacy of unconventional interventions in promoting student welfare and underscore the importance of considering innovative approaches to enhance well-being in educational settings.

Conclusion. Targeted Demand: Individuals diagnosed with AUD (84.3% aware, 65.7% likely/highly likely to use) and past naltrexone users (67.3% likely/highly likely to use, 90.4% likely/highly likely to recommend) showed the greatest interest in OTC access. Broad Acceptance: Across AUDIT-C groups and familiarity levels, over 80% anticipate a positive impact from OTC naltrexone and would recommend to others. More education needed: Concerns highlight the need for clear instructions due to lack of medical supervision in an OTC setting. In conclusion, this retrospective study provides compelling evidence that mammoth stuffed animals have a positive impact on student well-being. The presence of these endearing companions is associated with reduced stress levels, increased self-esteem, and a sense of emotional security. These findings support the integration of unconventional interventions to enhance student welfare in educational settings.

Neuroscience

Examining the effect of S-palmitoylation on $Na_V1.2$ localization and its interaction with ankyrins

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Purpose. Voltage-gated sodium channels (Na_Vs) are critical mediators of action potential generation and propagation in neurons. Improper localization of these channels impairs their function, and neuronal excitability, as a result. Na_Vs are anchored to specific excitable regions of the neuronal membrane by members of the ankyrin scaffolding protein family. Mutations of the genes encoding ankyrins and certain Na_Vs that disrupt localization have been implicated in a number of neurodevelopmental conditions. In particular, loss-of-function mutations of the *SCN2A* gene, which encodes the sodium channel $Na_V1.2$, represent one of the most prominent single-gene risk factors for autism spectrum disorder (ASD). The exact mechanism of interaction and localization between $Na_V1.2$ and its ankyrin binding partners has not yet been confirmed, but studies of other Na_V1 family members have identified the post-translational modification of S-palmitoylation as a potential regulator of Na_V/ankyrin binding. It has been shown that palmitoylation-deficient $Na_V1.2$ has altered localization early in development in cultured mouse neurons. It has also been shown that $Na_V1.6$ is palmitoylated at three distinct sites, two of which are located in the cytoplasmic loop (II-III loop) between the second and third homologous domains of the channel. This loop contains the ankyrin-binding domain, making it of particular importance.

Methods. Due to the conservation of cysteine residues across the Na_V1 family, we sought to create a mutant of the II-III loop of $Na_V1.2$ in which the cysteine sites were replaced with alanine residues to examine the effects of palmitoylation on $Na_V1.2$. Palmitoylation status of the wild type (WT) versus mutant II-III loop was assessed via the acyl biotin exchange (ABE) assay. Impacts of palmitoylation on the $Na_V1.2$/ankyrin complex were assessed via co-immunoprecipitation.

Results. We confirmed production of the mutant II-III loop via DNA gel and sequencing. We saw that the mutant II-III loop does not appear to undergo S-palmitoylation. We also saw interaction between the II-III loop and ankyrins B and G, although this occurred with both the WT and mutant loop.

Conclusion. Future studies will be conducted to optimize ABE and immunoprecipitation protocols so as to better assess palmitoylation-dependent interactions between the $Na_V1.2$ II-III loop and ankyrin binding partners and to explore the functional impacts of palmitoylation-dead $Na_V1.2$ in neurons.

Using GCamP6s Fluorescence in Zebrafish Hair Cells to Understand the Effects of Glutamate Agonist AMPA on Hair Cell Activity

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Purpose. Glutamate excitotoxicity refers to the neuronal damage and cell death associated with the accumulation of excess glutamate. Studies suggest that the accumulation of excess glutamate damages sensory hair cells, which are essential for hearing and balance. This study aimed to further assess the effects of alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA), a glutamate agonist, on hair cell activity in zebrafish. Because zebrafish hair cells activate differently to anterior-to-posterior ($A \rightarrow P$) and posterior-to-anterior ($P \rightarrow A$) fluid-jet stimulation, this study investigated the effects of AMPA on hair cell activity during $A \rightarrow P$ and $P \rightarrow A$ stimulation.

Methods. Using calcium imaging procedures with zebrafish, this study (1) validated hair cell responsiveness to $A \rightarrow P$ and $P \rightarrow A$ stimulation, (2) confirmed that hair cells respond differentially to $A \rightarrow P$ and $P \rightarrow A$ stimulation and classified hair cells as either $A \rightarrow P$ or $P \rightarrow A$ responding, and (3) collected baseline measures and monitored the effects of AMPA in 0.1% dimethyl sulfoxide (DMSO) in E3 over 50 minutes. .

Results. The results suggest that AMPA in 0.1% DMSO in E3 appears to inhibit hair cell activity during $A \rightarrow P$ stimulation for both $A \rightarrow P$ and $P \rightarrow A$ responding hair cells, but not during $P \rightarrow A$ stimulation for either group. Spontaneously, this study also discovered that 0.4% DMSO in E3 appears to completely inhibit general hair cell activity during $A \rightarrow P$ stimulation.

Conclusion. These results may suggest that AMPA affects hair cell activity in a direction-specific way. However, because 0.4% DMSO in E3 also inhibited hair cell activity, future studies must assess whether DMSO also inhibits hair cell activity at 0.1%, and whether it contributes to the effects of AMPA in 0.1% DMSO in E3. Studies should administer DMSO at 0.1% in E3 alone as well as AMPA in E3 without DMSO. Furthermore, both DMSO and AMPA groups must be compared to a pure control group with E3 only. A limitation of this study is that the $A \rightarrow P$ stimulation was stronger than the $P \rightarrow A$ stimulation. Future studies must assess whether the direction-specific effects of AMPA remain even after using similar strength of $A \rightarrow P$ and $P \rightarrow A$ stimulation.

Electrophysiological Diversity of Mouse Vestibular Ganglion Glia

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Purpose. In the peripheral nervous system, glial cells are responsible for the formation and maintenance of myelin, which is critical for signal propagation and neuronal survival. Beyond their role in myelination, little is known about the function of vestibular glia. Recent work in mouse spiral ganglion glia identified both inwardly and outwardly rectifying currents around the onset of hearing, suggesting a diversification of cochlear glial properties. Elsewhere in the nervous system, glia have also been shown to clear neurotransmitters, buffer potassium, and provide energy to neurons. As the vestibular ganglion and the spiral ganglion deliver different information from the ear to the brain, my work will investigate whether glia in the vestibular ganglion exhibit a similar electrophysiological diversity to those of the spiral ganglion.

Methods. Vestibular ganglia were excised from four C57BL/6 mice postnatal day 9 to 16. The ganglia were enzymatically treated, dissociated, and cultured for 1-2 nights to isolate single somata for patch clamp recordings. Recordings were made in the rupture-patch configuration to characterize whole-cell currents. To assess their voltage-activated membrane currents, glia were subjected to series of 200 ms voltage steps from -203 to +57 mV, in 10 mV increments, from a holding potential of -73 mV. Additionally, glial fibrillary acidic protein (GFAP) and $K_{ir}4.1$, an inwardly rectifying potassium channel, were detected using immunofluorescence in the vestibular ganglion of a postnatal day 29 mouse.

Results. Vestibular ganglion glia comprised distinct inwardly and outwardly rectifying cell populations, with inwardly rectifying currents being larger in amplitude. Of 26 successful glia with giga-ohm seals, 20/26 showed inwardly rectifying currents, and 6/26 showed outwardly rectifying currents. Glia with inwardly rectifying currents also had a more hyperpolarized resting membrane potential at -72.8 mV, close to the reversal potential of potassium. $K_{ir}4.1$ immunodetection showed strong fluorescence in the satellite glial cell layer but weak fluorescence in the Schwann cell layer.

Conclusion. In both the spiral ganglion and the vestibular ganglion, glia exhibit diverse membrane properties, indicating a divergence in function. For satellite glial cells, hypothesized to be the population of glia with inwardly rectifying currents, potassium clearance may be a critical role during the neuron's repolarization step. For Schwann cells, hypothesized to be the cells with outwardly rectifying currents, outward K_v currents have been reported to influence cell proliferation and differentiation. Mammalian Schwann cells have been shown to express K_{ir} only around the onset of myelination, which may indicate other mechanisms of potassium clearance following maturation.

Auditory Processing and Attention to Speech in Minimally Verbal Autism: A Pilot Study

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Purpose. Autism spectrum disorder (ASD) involves substantial heterogeneity in language ability and cognitive function. However, little is known about minimally verbal autism (MVA), which comprises an estimated 30% of the autistic population, as most research has focused on individuals who communicate through spoken language and demonstrate average to above-average intelligence. This pilot study examines how auditory processing and attentional orienting to novel auditory stimuli relate to communication and cognitive ability across minimally verbal autism (MVA), verbal autism (VA), and typically developing (TD) groups, aiming to uncover neurophysiological factors contributing to MVA.

Methods. Scalp EEG was recorded in 18 children ages 8–14 (MVA, n=6; VA, n=7; TD, n=5) during a passive auditory oddball paradigm presenting task-irrelevant tones and consonant-vowel (CV) sounds while participants watched a muted video. Standard and deviant stimuli belonged to the same stimulus class per block; novel stimuli were drawn from a different class. Participants also completed measures of receptive (PPVT) and expressive (EVT) language, cognitive ability (IQ), and adaptive behavior (Vineland). Mismatch negativity (MMN) and P3a components were derived from novel-minus-standard difference waves.

Results. Difference waves showed an early fronto-central deflection consistent with a MMN (~175 ms for tones, ~250 ms for CV sounds, though this classification is tentative for CV), followed by a P3a (~300 ms tone, ~340 ms CV). MVA participants showed a reduced P3a response to novel CV sounds relative to TD ($p=.013$, Holm-corrected), consistent with reduced automatic attentional orienting to unexpected speech sounds. Neural component amplitude showed preliminary associations with clinical measures of communication: tone MMN amplitude correlated negatively with Vineland-3 Communication scores ($r=-.55$, $p=.022$), and CV P3a amplitude correlated positively with Vineland-3 Communication scores ($r=.61$, $p=.009$).

Conclusion. These preliminary findings suggest that reduced attentional orienting to novel speech sounds, indexed by P3a, relates to communication ability across the autism spectrum. Data collection is ongoing to increase power for group comparisons and correlational analyses, and future analyses will extend to word-listening and continuous naturalistic-speech conditions to examine auditory processing under more naturalistic listening settings.

Characterization of Cam Kinase II Mutant Overexpression in Drosophila NMJ

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Purpose. The neuromuscular junctions (NMJ) of Drosophila share many similarities to humans and can be used as a model for studying neurological diseases. This study focuses on the impact of Calcium/Calmodulin-Stimulated Protein Kinase II (CaMKII) on synaptic development and function; laying the foundation for future work expanding on research conducted by UMass Amherst on the ability to reverse the effects of overexpression of CaMKII.

Methods. Three strains of driver flies were used: 1. OK371-Gal4 activates gene expression in glutamatergic neurons, 2. Elav-Gal4 is pan neuronal, activating gene expression in all post-mitotic neurons, or 3. C380 (X)-Gal4 activates gene expression in motor and some peripheral neurons. Each genetic cross contained three virgin females from one of the drivers and three males from either the control (CS), T287D mutant, or D136N, T287D mutant. In total, there were seven different crosses. Larvae from each of these crosses were dissected to expose the NMJ. Through immunohistochemistry using mouse α-BRP and rabbit α-GluRC as primary antibodies, the larvae were stained with fluorescent markers. The larvae were mounted onto a slide and imaged using a confocal microscope. The images were analyzed using Fiji to count the number of boutons, integrated density of BRP, integrated density of GluRC, and total area of the synapse. Ten larvae from each cross were used in a behavioral assay. Larvae were placed onto an agarose dish with graph paper underneath, and two trials of two minutes were performed to measure the total distance traveled. ANOVAs were used for statistical analysis using Prism.

Results. Flies with the T287D mutation inserted into both chromosome (Chr) II and III resulted in increased number of boutons at the NMJ. This occurred when using the OK371-Gal4 and elav-Gal4 driver. Additionally, larvae with the double point mutation (OK371-Gal4 x D136N, T287D) inserted into Chr III, had significantly less boutons than the T287D mutants while having no significant difference between the controls. The T287D mutation inserted into Chr II increased eclosion while when inserted into Chr III, decreased eclosion. There were no significant differences in the areas of the NMJs or locomotor behavior in larvae between crosses.

Conclusion. Proper expression of CaMKII is essential for NMJ development and function. The number of boutons increases as a result of the overexpression of CaMKII due to the insertion of UAS-T287D CaMKII. This indicates the increase in the number of boutons is a phosphorylation dependent mechanism rather than a localization or structural mechanism. In addition, elav-Gal4 which is a pan neuronal driver, impacts synaptic development (in terms of number of boutons) to a greater extent than more selective drivers (OK371-Gal4 and C380 (X)-Gal4).

Dopaminergic Signaling Contributes to the Regulation of Territorial Competition in Betta splendens

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Purpose. This study determines whether male Betta splendens develop a conditioned place preference (CPP) for a mirror-associated positive stimulus environment and the role of dopamine D1 receptor signaling in their aggressive behavior and locomotion.

Methods. A conditioned place preference (CPP) assay is a behavioral test used to evaluate the motivational and rewarding properties of a stimulus. In this study, male Betta splendens were exposed to either a mirror stimulus or a control condition for 10-15 minutes once daily over a 10-day conditioning period. Conditioned preference was assessed by measuring the amount of time fish spent on the stimulus-associated side of the testing tank. To investigate the role of dopamine signaling in aggression and associative learning, fish were treated with SCH, an antagonist drug that blocks dopamine activity, or SKF, an agonist drug that enhances dopamine activity. These tests were conducted to examine how changes in dopamine signaling influence Betta splendens aggressive behavior and locomotion.

Results. The results of the conditioned place preference (CPP) test suggested that there was no statistically significant correlation in the betta fish choosing between their initial preferred side (IPS) and the "reward" mirror stimulus. The data showed random learning habits, or lack thereof, amongst both control and mirror exposed fish. However, the results of the dopamine experiment indicated that there was a statistically significant difference between the latency to lateral display before and after the dopamine receptor activator (SKF) was administered to the fish. The data revealed that SKF significantly increased the latency after its application. These findings suggest that dopamine has some effect on competition and aggression.

Conclusion. The results from this study suggest that dopamine does in fact influence territorial competition by changing the Betta splendens' perception of social reward. These findings provide insight into how the neural mechanisms underlying aggression and reward may contribute to a broader understanding of how dopamine may affect social behavior across vertebrates.

Physics and Astronomy

Generating Morphology-Informed Multi-Sphere Representations of the LHS-1D Lunar Dust Simulant for DEM Simulations

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Purpose. Lunar regolith's fine, irregular, electrostatically charged particles threaten surface hardware and EVA suits. Discrete Element Method (DEM) simulations typically represent regolith as spheres, limiting fidelity to real contact and adhesion behavior. This work develops a morphology-informed workflow that converts measured LHS-1D lunar dust simulant morphology into non-spherical, DEM-compatible multi-sphere particles, supporting future simulations of dust adhesion, transport, and mitigation strategies.

Methods. Principal Component Analysis extracted size and shape descriptors (equivalent diameter, aspect/elongation/flatness indices, sphericity, roughness) from over 71,000 LHS-1D STL meshes in the NIST particle database. These distributions parameterized a Spherical Harmonics Particle Shape Generator, producing 10,000 synthetic particles, with low-degree terms setting base shape and higher-degree terms adding irregularity and roughness. Two workflows then converted particles into DEM-ready multi-sphere clumps: an STL-reference method fitting spheres to each generated mesh, and a Direct method sampling morphology statistics without an intermediate mesh. Clumps were imported into LIGGGHTS to verify DEM compatibility.

Results. Generated particles reproduced equivalent diameter, aspect ratio, elongation, and sphericity with close-to-moderate agreement (mean errors under 6%, SD ratios 0.84–0.98), while roughness agreement was poor (40% mean error). Among clump-fitting workflows, the Direct method matched bulk shape descriptors better than the STL-reference method and generated 1,000 clumps roughly 172× faster (0.183 s vs. 31 s per particle, two-worker configuration), though both showed poor roughness agreement (19–94% mean error). Form classification confirmed particles were predominantly compact, consistent with the measured LHS-1D population. Successful LIGGGHTS import confirmed correct clump reconstruction.

Conclusion. This workflow links measured LHS-1D morphology to computationally efficient, non-spherical DEM particles, with the Direct method offering the best balance of speed and fidelity for large populations. Surface roughness remains the primary limitation and target for refinement. Future work will integrate a screened Coulomb–Yukawa electrostatic solver into LIGGGHTS to study charge-driven dust adhesion and transport relevant to Artemis surface operations.

Swendsen-Wang Algorithm And The Invaded Cluster Algorithm

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Purpose. A fridge magnet suddenly losing its magnetism after reaching a certain temperature is an example of a phase transition, where a system goes from one state to another state instantly. Phase transitions are very important for understanding some of the phenomena in the world. To better understand them, we can use the 2D Ising Model to model certain systems such as ferromagnets with phase transitions and analyze its behavior. One of the challenges is finding an appropriate algorithm to simulate the random changes in the system. In this study, we will look at cluster algorithms to see its effectiveness and efficiency in modeling phase transitions with the 2D Ising Model.

Methods. For this study, we will be modeling the magnetization of a flat ferromagnet with the 2D Ising Model. The ferromagnet will be modeled with a 2D lattice of spin states, where each spin state is either up $(+1)$ or down (-1). To simulate the randomness of the system, we use a cluster algorithm called the Swendsen-Wang Algorithm, where clusters are formed and flipped with a probability dependent on the temperature of the system. We measured various observables such as magnetization and compared this data with the Metropolis Algorithm data. We also used the Invaded Cluster Algorithm to calculate the critical temperature at which the phase transition occurs.

Results. We found that the Swendsen-Wang Algorithm does a great job at modeling the phase transition of the ferromagnet. In addition, we examined the equilibration and autocorrelation times of the Swendsen-Wang Algorithm, and we found that these times were significantly less at the critical temperature for the Swendsen-Wang Algorithm than the times for the Metropolis Algorithm. This demonstrates that the Swendsen-Wang Algorithm is not as affected by the effects of critical-slowing down compared to the Metropolis Algorithm. Also, our calculated value for the critical temperature by the Invaded Cluster Algorithm is 2.8736 ± 0.0668 , which is far off from the theoretical value of 2.2692. This is likely due to the smaller lattice sizes used to carry out the measurements (which was done due to long runtime issues).

Conclusion. We found that the Swendsen-Wang Algorithm effectively models the 2D Ising Model and significantly reduces the runtime of the simulations near the critical temperature. Also, the measurement of the critical temperature via the Invaded Cluster Algorithm can be significantly improved by optimizing the runtime of the algorithm for higher lattice sizes.

Brane trajectories in cosmological spacetimes

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Purpose. A fundamental mystery of our universe is its particular rate of expansion, fueled by an outward pressure known as dark energy. While we can observe the impacts of dark energy on our universe, we cannot experimentally compare different universes to determine the causes or consequences of a specific dark energy content. Instead, we use theoretical methods to better understand the role of dark energy in cosmology through describing the behavior of objects in 3+1-dimensional spacetimes.

Methods. We apply mathematical tools from general relativity, differential geometry, and string theory to study the trajectories of general-dimensional objects moving in spacetimes, known as branes. By applying the variational principle to brane area-action, we derive the equation of motion for a general brane which is dependent on its geometry. We then apply this formalism to compare the trajectories of massive, spherical 2-branes in Minkowski and de Sitter spacetimes.

Results. The equations of motion for a massive, spherical 2-brane in Minkowski spacetime demonstrate that it tends toward shrinking, regardless of initial conditions. By contrast, a massive, spherical 2-brane in de Sitter spacetime has a critical radius at which the inward surface tension and outward pressure perfectly balance. When the brane is larger than the critical radius, it tends toward expanding as the outward force due to dark energy is greater than inward force from surface tension. Correspondingly, when the brane is smaller than the critical radius, it tends toward shrinking. These results align with their classical analogue, where higher pressure within a bubble allows it to hold its form, while no pressure difference would lead to a shrinking bubble.

Conclusion. While the dark energy content of a spacetime is conventionally seen as a constant, branes with negative energy can be absorbed by black holes, leading to a change in the dark energy content of the spacetime. This research opens the door to further study of mechanisms for changing the dark energy content of spacetimes via black hole energy extraction. Our eventual goal is to describe negative-energy charged branes absorbed by a rotating Bañados-Teitelboim-Zanelli (BTZ) black hole, as a model mechanism enabling the fine-tuning of our universe's dark energy content.

Investigation of Student Experience with Computational Physics and Computational Astronomy Activities

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Purpose. Modern day physics and astronomy research has increasingly relied on computational methods for solving complex problems. Accordingly, the Next Generation Science Standards has recognized computation as a priority for physics instruction. The Physics Education Research (PER) community has thought extensively about course design in introductory and upper-division physics, but computation in undergraduate research and specific practices have been understudied. We aim to examine students' experience with coding and programming in selected courses and undergraduate research. We ask: 1) How do physics/astronomy students from different levels of research and coding approach computational learning and problem solving? 2) How do introductory and self-explanatory materials can aid the overall learning experience of students in programming?

Methods. End-of-Spring 2026 surveys were collected from 18 students: 10 from PHYS 125: Oscillations and Waves, two from ASTR 200: Intro to Data Science with Astronomical Applications, and six Physics and Astronomy thesis students. We conducted a 5-step thematic analysis using Notion and Dedoose: (1) reviewing a subset of transcripts, (2) summarizing responses, (3) generating initial codes, (4) applying and refining the coding scheme across the full dataset, and (5) grouping codes into overarching themes and inter-Rater Reliability was conducted to achieve a higher percentage of agreement.

Results. Through thematic analysis, we find 6 prominent themes across our data in students' experience with programming. Comparing beginner and research students (RQ1): We find that beginners see coding as less connected to their content learning/careers goals while research students see it as integral, though both groups value Python's data visualization capabilities. Research students also plan and structure their code more deliberately than beginners, and research students and beginner students have different debugging processes in terms of AI-usage and optimization. Examining introductory materials, we see (RQ2): current Python materials in PHYS 125 can be aided with more deliberate design for supporting students with various coding experience and background.

Conclusion. Following our thematic analysis, we propose four recommendations for future course improvement to Incorporate large-scale projects, differentiate the intro file by skill level, provide TAs with preparation resources, and expand debugging-focused office hours. Possible limitations to our research include our small sample size and survey structure. In the future, we could refine the questions on the survey to better capture student experiences and expand to a larger sample size.

SnO_2-Coated AFM Tips for Nanoscale Electrical Characterization of Perovskite Solar Cells

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Purpose. Currently, commercial solar devices do not exceed 30% power conversion efficiency (PCE). Perovskites present a valuable alternative to traditional silicon solar cells. The tunable bandgap, energy conversion efficiency, and quick energy payback of perovskites are inherent to their unique crystalline structure. Improving perovskite solar cell technology requires direct analysis of the perovskite layer at the Electron Transport Layer (ETL) interface. This interface is analyzed using an Atomic Force Microscope (AFM), which moves a sharp probe tip across a surface to map topography and local conductivity at the nanoscale. This research aims to improve perovskite analysis and effectively increase PCE, actualizing solar's potential in the renewable energy landscape.

Methods. Various methods were explored for depositing SnO_2, a common perovskite ETL, onto microscopic AFM probe tips. Film thickness, stoichiometry, and electron mobility were optimized. Although not viable for the AFM probe itself, spin coating is a standardized technique that was used as a target for feasible tip deposition methods. Substrates were coated to establish parameters that were then applied to AFM tips. Tested methods included sputtering, where deposition time and oxygen concentration were optimized, and dip coating, where withdrawal rate and solution concentration were adjusted. Thin films were characterized using AFM and UV-vis spectroscopy. Additionally, bare tips on coated substrates were compared to coated tips on bare substrates.

Results. Analysis revealed that increasing oxygen concentration during sputtering yielded a bandgap that approached the spin coated thin film. Meanwhile, dip coated substrates exhibited bandgaps and electrical characteristics more closely aligned with spin coating. Electrical characterization of SnO_2-coated AFM tips showed that ideal substrate deposition conditions are not directly transferable to ideal deposition conditions for tips. This mismatch in electrical behavior is likely tip films are significantly thinner than substrates films. Ultimately, the sputtered tips exhibited more uniform coatings than the dip coated ones.

Conclusion. Although dip coating naturally yields a bandgap and electrical characteristics closer to spin coating, sputtering has greater potential due to its tunability and also results in a compact layer that has not yet been achieved through dip coating. Future work will optimize sputtering for improved electrical properties, refine dip coating for sharp tip geometry, and compare coated versus bare tips on perovskite. These findings provide a foundation for advancing nanoscale characterization and realizing high-efficiency perovskite solar cells.

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The Effect of Outer Giant Planets on Inner System Architectures

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Purpose. Understanding the processes that dictate planetary formation and evolution is crucial to constraining how the architectures of exoplanetary systems emerge, as well as for determining regimes of stability and chaos. Observations from NASA's Kepler mission revealed a remarkable trend of intra-system uniformity in inner terrestrial planet masses, radii, and orbital spacings, a phenomenon referred to as "peas in a pod" architectures. Evidence suggests that these system architectures may result from later dynamical sculpting, rather than being a natural outcome of early planet formation. Furthermore, systems featuring an orbital gap between two of the inner compact planets preferentially harbor an outer gas giant, suggesting a possible dynamical process whereby the outer giant induces instabilities, collisions, and ejections that reshape inner system architectures.

Methods. Here, we investigate whether the inward migration of an outer giant planet can sweep strong enough gravitational resonances to adequately pump the eccentricities and inclinations of inner planets to generate the observed gaps. Using the N-body integrator REBOUND, we simulate randomized inner compact terrestrial systems and track their evolution as a Jupiter-mass planet migrates inward. We complement these numerical simulations with analytical Laplace-Lagrange secular theory calculations using celmech to evaluate mean motion resonances and secular resonance crossings over time.

Results. Preliminary results demonstrate that a single outer giant is too weakly coupled to the terrestrial inner planets to sweep strong secular resonances. Moreover, significant differences in semi-major axes between the outermost terrestrial planet and the giant planet ensure that only high-order mean motion resonances are encountered during migration, which are too weak to trigger dynamical sculpting.

Conclusion. Thus, our early results suggest that a second giant planet may be required to sweep resonances more strongly, or that another unaccounted-for dynamical process is shaping the system architectures we observe.